Python Cheatsheet

MATH60033A · Session 08 · KNN and bias–variance

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Everything you need to type, in one place. The first three sections are the same every week; the last one is the Python from session 7.

Starting Python in VS Codium

Open the course folder once — **File > Open Folder**, then pick the repository. Everything below assumes you are inside it.

Open a terminal with **View > Terminal** (`Ctrl` + ` on Windows and Linux, `Cmd` + ` on macOS). Create the environment the first time only:

```
python3 -m venv .venv       # macOS
python3 -m venv .venv       # Linux
py -m venv .venv            # Windows (PowerShell)
```

Then activate it — **every time you open a new terminal**:

```
source .venv/bin/activate  # macOS
source .venv/bin/activate  # Linux
.venv\Scripts\activate     # Windows (PowerShell)
```

You will see `(.venv)` at the start of the prompt. If you do not, nothing below will find pandas. Install the packages once, with the environment active:

```
pip install -r requirements.txt
```

Finally tell the editor which Python to use: `Ctrl/Cmd` + `Shift` + `P`, type **Python: Select Interpreter**, and choose the one inside `.venv`.

A new file, and running it

File > New File, then save it with a `.py` ending — `session03.py`, say. Write your code, save it, and in the terminal:

```
python session03.py         # macOS, Linux and Windows alike, inside .venv
```

Inside the environment the command is `python` on all three platforms. Outside it, macOS and Linux need `python3` and Windows needs `py`, which is one more reason to activate first.

For trying one line at a time, type `python` on its own to get the `>>>` prompt, and `exit()` to leave. The editor's **Run** triangle does the same thing as the command above.

The four things you always do

```python
import pandas as pd

# 1 - build a DataFrame
df = pd.DataFrame({"country": ["FR", "DE", "IT"],
                   "gdp_pc_eur": [37_000, 46_000, 32_000]})

# 2 - look at what is in it
df.head()           # the first five rows
df.shape            # (rows, columns)
df.dtypes           # what type each column holds
```

```python
df.info()          # types, and how many values are missing
df.describe()      # count, mean, sd and quartiles, per numeric column

# 3 - read a file in
df = pd.read_csv("data.csv")
df = pd.read_excel("data.xlsx")                   # needs openpyxl
df = pd.read_parquet("data/spine/core.parquet")

import qmib
df = qmib.load("core")                            # a course dataset, by name

# 4 - write a CSV out
df.to_csv("out.csv", index=False)   # index=False, or you get a stray column
```

From session 7

Describe

```python
np.linalg.eigvalsh(np.corrcoef(Z, rowvar=False))
```

Fit

```python
from sklearn.decomposition import PCA
p = PCA().fit(Z)                  # on standardised columns

import prince
famd = prince.FAMD(n_components=3).fit(mixed)   # numeric AND categorical

from statsmodels.multivariate.factor import Factor
f = Factor(df.to_numpy(), n_factor=2, method="ml").fit()
```

Read the fit

```python
p.explained_variance_ratio_       # what the scree plot is drawn from

p.components_                     # loadings: variables on components

f.rotate("varimax")               # same fit, readable loadings
```